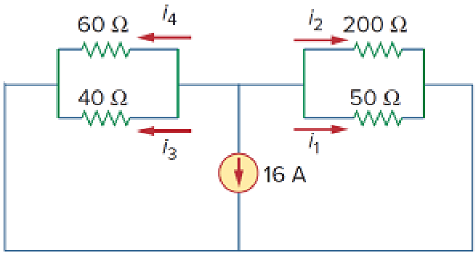
Chapter 2, Problem 32P

Problem

Find *i*1 through *i*4 in the circuit in Fig. 2.96.

Figure 2.96:



Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

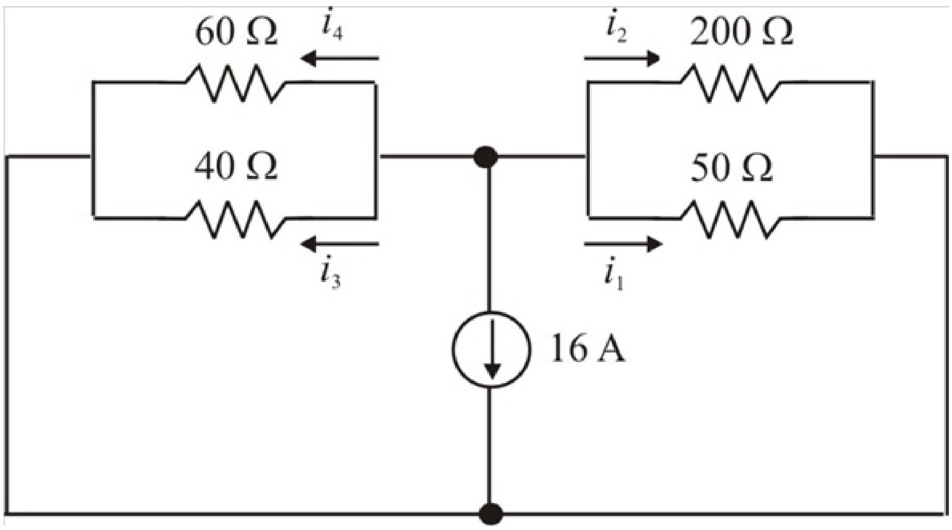


Figure 1

Step 2 of 8

The 200 ? and 50 ? resistors are in parallel. Their combined resistance is,

$$200\,\Omega \,||\, 50\,\Omega = \frac{200 \times 50}{200+50}$$
$$= 40\,\Omega$$

The 60 ? and 40 ? resistors are in parallel. Their combined resistance is,

$$60\,\Omega \,||\, 40\,\Omega = \frac{60 \times 40}{60+40}$$
$$= 24\,\Omega$$

All the currents are leaving the node. By changing the current source the arrow upwards, the current is -16 A.

Step 3 of 8

Then the reduced circuit is shown in Figure 2.

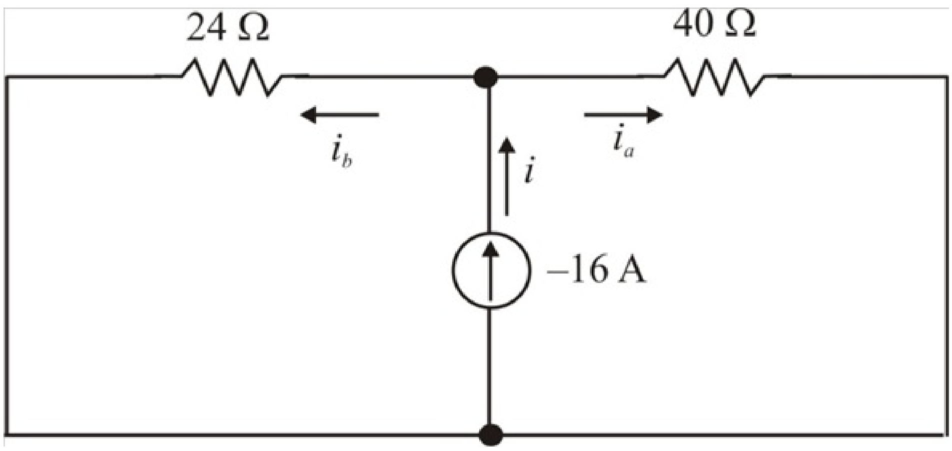


Figure 2

Step 4 of 8

Apply current division to the circuit in Figure 2.

The current passing through right is,

$$\begin{aligned} i_a &= \frac{24}{24+40} i \\ &= \frac{24}{24+40}(-16) \\ &= -6\,\text{A} \end{aligned}$$

The current passing through left is,

$$\begin{aligned} i_b &= \frac{40}{40+24} i \\ &= \frac{40}{40+24}(-16) \\ &= -10\,\text{A} \end{aligned}$$

Step 5 of 8

Then the circuit is shown in Figure 3.

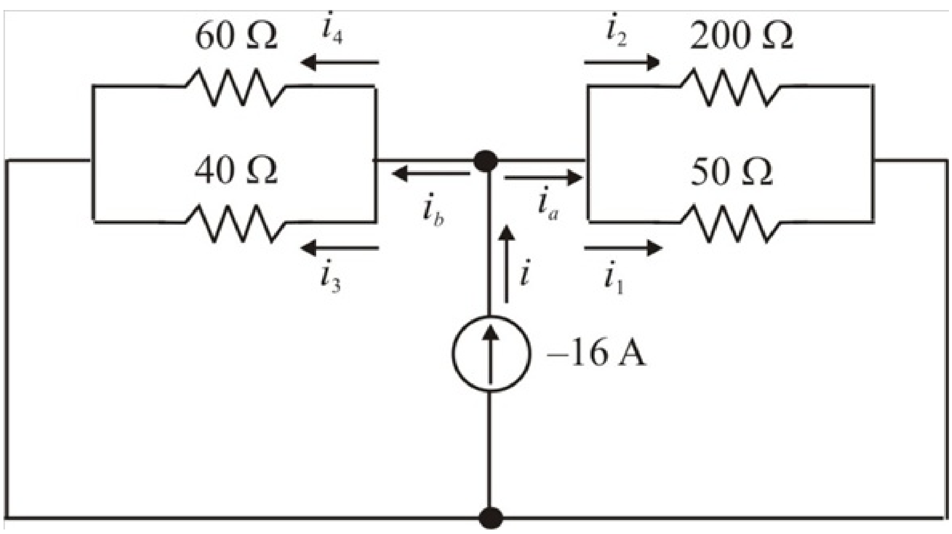


Figure 3

Step 6 of 8

By using current division, the current i_1 can be calculated as follows.

$$\begin{aligned} i_1 &= \frac{200}{200+50} i_a \\ &= \frac{200}{200+50}(-6) \\ &= -4.8\,\text{A} \end{aligned}$$

Therefore, the value of current i_1 is $\boxed{-4.8\,\text{A}}$.

By using current division, the current i_2 can be calculated as follows.

$$\begin{aligned} i_2 &= \frac{50}{50+200} i_a \\ &= \frac{50}{50+200}(-6) \\ &= -1.2\,\text{A} \end{aligned}$$

Therefore, the value of current i_2 is $\boxed{-1.2\,\text{A}}$.

Step 7 of 8

By using current division, the current i_3 can be calculated as follows.

$$\begin{aligned} i_3 &= \frac{60}{60+40} i_b \\ &= \frac{60}{60+40}(-10) \\ &= -6\,\text{A} \end{aligned}$$

Therefore, the value of current i_3 is $\boxed{-6\,\text{A}}$.

Step 8 of 8

By using current division, the current i_4 can be calculated as follows.

$$\begin{aligned} i_4 &= \frac{40}{40+60} i_b \\ &= \frac{40}{40+60}(-10) \\ &= -4\,\text{A} \end{aligned}$$

Therefore, the value of current i_4 is $\boxed{-4\,\text{A}}$.